Overview

4 Stratified simple random sampling

4.1 Estimation from sample data

4.2 Statistical inference

4.3 Sample size allocation

4.4 Efficiency gains from stratification

4.5 When and how to stratify?

4 Stratified simple random sampling

- The population U is **partitioned** into H mutually exclusive (and exhaustive), nonempty **strata** (subpopulations) U_h $(h = 1, \ldots, H)$.

- Stratum U_h contains N_h elements.

- It follows that

$$U = \bigcup_{h=1}^{H} U_h \qquad \text{and} \qquad N = \sum_{h=1}^{H} N_h,$$

  or, in short notation, $U = \bigcup_h U_h$ and $N = \sum_h N_h$.

- The partition **assigns** each element $i \in U$ to exactly one stratum.

Table: Population of farms in a U.S. county

Stratum	Farm area (grouped)	Number of farms	Mean corn acreage	Standard deviation
1	0 – 40	394	5.4	8.3
2	41 – 80	461	16.3	13.3
3	81 – 120	391	24.3	15.1
4	121 – 160	334	34.5	19.8
5	161 – 200	169	42.1	24.5
6	201 – 240	113	50.1	26.0
7	241 –	148	63.8	35.2
Mean or total		2 010	26.3	–

Source: Cochran (1977, p. 111); areas in acres (1 acre ≈ 0.4 hectares).

- Stratification **primarily** serves **two purposes**:
  - **Efficiency:** more precise estimates than under SRS through suitable stratification and sample size allocation
  - **Estimation for planned domains:** subpopulations are defined as *planned* domains with predetermined sample sizes
- In practice, the allocation may also account for stratum-specific characteristics such as variability or anticipated response rates (Chapter 4.5).
- The **notation** is somewhat **cumbersome**. We therefore introduce the main notation for totals at this point.
- Values of the y-variable in stratum h are identified using the **double subscript** ih and written as y_{ih}.

- The **population total** of the y-variable **in stratum** h is denoted by $t_{y,h}$ $(h = 1, \ldots, H)$. We write:

$$
\begin{aligned}
N_h &= \text{size of stratum } h,r \\
N &= \sum_{h=1}^{H} N_h \; = \text{ population size,} \\
t_{y,h} &= \sum_{i \in U_h} y_{ih} \; = \; y\text{-total in stratum } h, \\
t_y &= \sum_{h=1}^{H} t_{y,h} \; = \; y\text{-total in the population.}
\end{aligned}
$$

- The **population total** in can equivalently be written as

$$
t_y = \sum_{h=1}^{H} N_h \, \bar{y}_{h,U},
$$

where $\bar{y}_{h,U}$ is the **mean** in stratum h, given by

$$\bar{y}_{h,U} = \frac{1}{N_h} \sum_{i \in U_h} y_{ih} \qquad (h = 1, \ldots, H).$$

- The **population mean**, denoted by $\bar{y}_U$, is the weighted sum of the stratum means $\bar{y}_{h,U}$,

$$\bar{y}_U = \sum_{h=1}^{H} W_h \, \bar{y}_{h,U},$$

where

$$W_h = \frac{N_h}{N} \qquad (h = 1, \ldots, H)$$

is a **stratum weight**.

- The **variance** of the y-variable in stratum h is given by

$$\sigma_h^2 = \frac{1}{N_h - 1} \sum_{i \in U_h} (y_{ih} - \bar{y}_{h,U})^2 \qquad (h = 1, \ldots, H).$$

- The **population variance** is

$$\sigma_y^2 = \frac{1}{N - 1} \sum_{h=1}^{H} \sum_{i \in U_h} (y_{ih} - \bar{y}_U)^2.$$

Aside (One-way analysis of variance).

An alternative expression for the variance σ_y^2 is

$$\sigma_y^2 = \frac{SST}{N-1},$$

where **SST** denotes the total sum of squares. It can be decomposed into two components: the sum of squares within strata (**SSW**) and the sum of squares between strata (**SSB**); formally,

$$SST = SSW + SSB,$$

with

$$SSW = \sum_{h=1}^{H} (N_h - 1)\sigma_h^2 \quad \text{and} \quad SSB = \sum_{h=1}^{H} N_h \left(\bar{y}_{h,U} - \bar{y}_U\right)^2.$$

4.1 Estimation from sample data

- A **stratified** probability sample has the following **characteristics**:
  - Each stratum U_h is treated as a **separate subpopulation**.
  - A **separate**, **independent sample** s_h of size n_h is selected from each stratum U_h $(h = 1, \ldots, H)$.
- Independence implies that
  - a **different** sampling design can, in principle, be used in each stratum;
  - the **variance** of an estimator can be **decomposed** (in short notation) as

$$\text{var}_p = \sum_h \text{var}_{p,h}$$

- For **practical reasons**, it is advisable to use designs from a single family (e.g., SRS), particularly to keep variance estimation manageable.
- We **restrict attention** to the case in which an **SRS** is selected independently within each stratum.
- The **sample sizes** n_h $(h = 1, \ldots, H)$ can be allocated across strata in **different ways** (e.g., proportional to N_h or according to another rule).
- The stratified design is denoted by p_{STSI}.
- The **overall sample** s and its size n are obtained as the **union** of the stratum samples s_h, $h = 1, \ldots, H$, and the sum of their sample sizes n_h,

$$s = \bigcup_{h=1}^{H} s_h \qquad \text{and} \qquad n = \sum_{h=1}^{H} n_h.$$

- Let s be a sample under design p_{STSI}. The estimators of the **mean** and **variance** in stratum h are

$$\bar{y}_h = \frac{1}{n_h} \sum_{i \in s_h} y_{ih} \qquad \text{and} \qquad \widehat{\sigma}_h^2 = \frac{1}{n_h - 1} \sum_{i \in s_h} (y_{ih} - \bar{y}_h)^2 \qquad (h = 1, \ldots, H),$$

  - **provided** that $n_h > 0$ (mean) and $n_h > 1$ (variance), respectively.
  - Both estimators are unbiased for the corresponding population characteristics.

- An unbiased **estimator** of the **population mean** is

$$\bar{y}^{\mathrm{strat}} = \sum_{h=1}^{H} W_h \, \bar{y}_h.$$

- The theoretical **variance** of $\bar{y}^{\text{strat}}$ is

$$\text{var}_p(\bar{y}^{\text{strat}}) = \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h}. \tag{4.2}$$

- An unbiased **variance estimator** is

$$\widehat{\text{var}}_p(\bar{y}^{\text{strat}}) = \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\widehat{\sigma}_h^2}{n_h}.$$

- An unbiased estimator of the **population total** t_y is

$$\widehat{t}^{\,\text{strat}} = \sum_{h=1}^{H} N_h \, \bar{y}_h \qquad \text{or} \qquad \widehat{t}^{\,\text{strat}} = N \, \bar{y}^{\text{strat}}.$$

- The **theoretical** variance of $\widehat{t}^{\,\mathrm{strat}}$ is

$$\mathsf{var}_p\big(\widehat{t}^{\,\mathrm{strat}}\big) = N^2\,\mathsf{var}_p\big(\bar{y}^{\mathrm{strat}}\big).$$

- An unbiased **variance estimator** is

$$N^2\,\widehat{\mathsf{var}}_p\big(\bar{y}^{\mathrm{strat}}\big).$$

4.2 Statistical inference

- The design p_{STSI} of the overall sample consists of H stratum-specific designs, **all based on SRS**.
- It is therefore natural to construct **confidence intervals analogously** to those under SRS (Chapter 3).
- There are **two asymptotic settings**:
  1. The number of strata H is fixed and $n_h, N_h \to \infty$.
  2. The stratum sizes n_h and N_h are fixed and $H \to \infty$.
- A **central limit theorem** can be established in both settings; see Krewski and Rao (1981).

- For sufficiently large sample sizes, an approximate $100(1-\alpha)\%$ **confidence interval** (CI) for $\bar{y}_U$ is

$$\bar{y}^{\text{strat}} \pm z_{1-\alpha/2}\, \widehat{\text{SE}}_p(\bar{y}^{\text{strat}}),$$

where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the $N(0,1)$ distribution.

  - As under SRS, this confidence interval is based on a **normal approximation**.
  - Its accuracy depends on the finite populations within the strata and on the stratum sample sizes n_h, in particular when distributions are skewed.
  - Since sampling within each stratum is SRS, **Cochran's rule** from Chapter 3 can be used as a **rough guide within individual strata**.

4.3 Sample size allocation

- Let n denote the **predetermined** overall sample size.
- When **planning** the survey, we must decide how to **allocate** n across the strata such that

$$\sum_{h=1}^{H} n_h = n$$

holds.

- We consider two methods for **allocating** the total sample size n across strata:
  - **Proportional** allocation
  - **Neyman–Tschuprow** (optimal) allocation

Proportional allocation

- Proportional allocation can be used when **no information beyond** the stratum population sizes $N_1, \ldots, N_H$ is available.
- Under this allocation, the sample sizes n_h are chosen **proportional** to N_h $(h = 1, \ldots, H)$.
- The stratum sample sizes are set according to

$$n_h = \left[n \frac{N_h}{N} \right] = [n W_h], \quad \text{with} \quad W_h = \frac{N_h}{N} \qquad (h = 1, \ldots, H),$$

  where $[x]$ denotes a **rounding function** that rounds x to an integer.
- Because the n_h must be integers, rounding may result in $\sum_{h=1}^{H} n_h \neq n$. The rounded values must therefore be adjusted if the equality is to hold.

- The general **variance formula** for the population mean under a stratified design in (4.2) simplifies for the **strategy** $(p^{\text{prop}}, \bar{y}^{\text{strat}})$ to

$$\text{var}_p^{\text{prop}}(\bar{y}^{\text{strat}}) = (1-f)\frac{1}{n}\sum_{h=1}^{H} W_h \sigma_h^2.$$

- We use the short notation p^{prop} instead of $p^{\text{prop}}_{\text{STSI}}$.

Example 13 (Corn acreage).

A stratified SRS of size $n = 100$ is to be selected from U (see Table 1) using proportional allocation. The resulting stratum sample sizes n_h are shown below.

Example (Cont.).

Stratum h	1	2	3	4	5	6	7
$W_h = N_h/N$	0.196	0.229	0.195	0.166	0.084	0.056	0.074
$n_h = [nW_h]$	20	23	19	17	8	6	7

Optimal allocation according to Neyman–Tschuprow

- The allocation proposed by Neyman (1934) and Tschuprow (see Dalenius, 1992) assumes that, **in addition** to the stratum population sizes $N_1, \ldots, N_H$, the **stratum-specific population variances** of the variable of interest, $\sigma_1^2, \ldots, \sigma_H^2$, are known.

- Under this allocation, the stratum sample sizes are set as follows:

$$n_h = \left[n \frac{W_h \sigma_h}{\sum_{j=1}^{H} W_j \sigma_j} \right], \quad \text{with} \quad W_h = \frac{N_h}{N} \qquad (h = 1, \ldots, H), \tag{4.3}$$

  where $[\cdot]$ denotes the rounding function.

- The allocation is **variance-optimal** because it minimizes the variance in (4.2) subject to the constraint

$$\sum_{h=1}^{H} n_h = n.$$

- The sampling design is denoted by p^{opt} for short (full notation: $p^{\text{opt}}_{\text{STSI}}$).

- The **variance** of the estimator $\bar{y}^{\text{strat}}$ under the strategy $(p^{\text{opt}}, \bar{y}^{\text{strat}})$ is

$$\text{var}_p^{\text{opt}}(\bar{y}^{\text{strat}}) = \frac{1}{n}\left(\sum_{h=1}^{H} W_h \sigma_h\right)^2 - \frac{1}{N}\sum_{h=1}^{H} W_h \sigma_h^2.$$

- The Neyman–Tschuprow allocation has the following **properties**:
  - Strata with **large population sizes** N_h tend to receive **larger** sample sizes n_h.
  - Strata with **large population variances** σ_h^2 tend to receive **larger** sample sizes n_h.
  - If $\sigma_1 = \sigma_2 = \ldots = \sigma_H$, the allocation reduces to proportional allocation.

- In practice:
  - The **population standard deviations** σ_h $(h = 1, \ldots, H)$ are usually unknown. Estimates $\widehat{\sigma}_h$ may be substituted for σ_h in (4.3).
  - **Estimates** may be obtained, for example, from **historical data** or **related surveys**;
  - Surveys usually collect many variables, whereas Neyman–Tschuprow allocation is defined for a single variable. It may therefore be useful to compute allocations for several variables of interest and combine them into a **"compromise allocation."**
- If unit sampling costs differ across strata, the **cost-optimal allocation** also depends on the stratum-specific costs.

Example 14 (Corn acreage, cont. of Ex. 13).

A stratified SRS of size $n = 100$ is to be selected from U (see Table 1) using variance-optimal allocation. The resulting stratum sample sizes n_h are shown below.

Stratum h	$W_h = N_h/N$	$W_h\sigma_h$	Share	n_h
1	0.196	1.627	0.096	10
2	0.229	3.046	0.179	18
3	0.194	2.929	0.172	17
4	0.166	3.287	0.193	19
5	0.084	2.058	0.121	12
6	0.056	1.456	0.086	9
7	0.074	2.605	0.153	15
Total			1.000	100

4.4 Efficiency gains from stratification

- Strategies based on stratified sampling designs can, under certain conditions, achieve **efficiency gains** relative to SRS.
- We speak of an efficiency gain when an estimator has a **smaller variance than the reference estimator**.
- All sampling designs considered have the **same sample size** n.
- The **reference estimator** is $\bar{y}_{\mathrm{SRS}} := \bar{y}$, corresponding to the strategy $(p_{\mathrm{SRS}}, \bar{y}_{\mathrm{SRS}})$ with variance $\mathrm{var}_p^{\mathrm{SRS}}$.
- As before, $\mathrm{var}_p^{\mathrm{prop}}$ and $\mathrm{var}_p^{\mathrm{opt}}$ denote the variances under STSI with proportional and optimal allocation, respectively.

Proportional allocation vs. SRS

- The **efficiency gain** of the strategy $(p^{\text{prop}}, \bar{y}^{\text{strat}})$ over $(p_{\text{SRS}}, \bar{y}_{\text{SRS}})$ can be expressed as the **difference in variances** (in shorthand notation),

$$\text{var}_p^{\text{SRS}} - \text{var}_p^{\text{prop}} = \frac{1-f}{n(N-1)} \left\{ SSB - \sum_{h=1}^{H} (1-W_h)\sigma_h^2 \right\}.$$

$\Rightarrow$ The strategy $(p^{\text{prop}}, \bar{y}^{\text{strat}})$ is **more efficient** than $(p_{\text{SRS}}, \bar{y}_{\text{SRS}})$ if

$$\sum_{h=1}^{H} (1-W_h)\sigma_h^2 < SSB.$$

$\Rightarrow$ Thus, proportional stratification is more efficient than SRS when **between-stratum variation** is sufficiently large relative to **within-stratum variation**.

- The **SSB** is **maximized** when the strata are formed so that the stratum-specific **means** $\bar{y}_{h,U}$:
  - are as **different** as possible across strata,
  - while the strata are as **homogeneous** as possible internally.

Optimal vs. proportional allocation

- The **efficiency gain** of the strategy $(p^{\text{opt}}, \bar{y}^{\text{strat}})$ over $(p^{\text{prop}}, \bar{y}^{\text{strat}})$ can be expressed as the following difference in variances:

$$\text{var}_p^{\text{prop}} - \text{var}_p^{\text{opt}} = \frac{1}{n} \sum_{h=1}^{H} W_h (\sigma_h - \bar{\sigma})^2,$$

where $\bar{\sigma} = \sum_{h=1}^{H} W_h \sigma_h$ is the **weighted mean of the standard deviations** σ_h.

Example 15 (Corn acreage, cont. of Ex. 13 and 14).

For the population of farms in Table 1, we compare two strategies for estimating the population mean corn acreage: stratified SRS with proportional allocation and with variance-optimal allocation. The estimator variances (rounded to two decimal places) are $\text{var}_p(\bar{y}^{\text{prop}}) = 3.26$ and $\text{var}_p(\bar{y}^{\text{strat}}) = 2.73$. Thus, the strategy $(p^{\text{opt}}, \bar{y}^{\text{strat}})$ has a theoretical variance that is about 16% smaller.

4.5 When and how to stratify?

Reasons for using stratified sampling include:

- **Efficiency considerations:**
  - With suitable stratification and sample size allocation, a stratified sample can yield **more precise** estimates.
  - Gains in precision may arise when strata differ in their **means** or **variances** and the allocation exploits these differences.
  - In particular, stratification is effective when strata are internally **homogeneous** with respect to the variable of interest.
- **Estimation for subpopulations:**
  - Stratification allows subpopulations to be defined as **planned domains** and their sample sizes to be set to achieve the required precision.

- **Administrative convenience:** A national survey may, for example, be easier to organize if each region (e.g., canton) independently collects its assigned part of the sample. In this case, regions may provide a suitable stratification.

- **"Balanced" samples (control of sample composition):**
  - A stratified sample can prevent a potentially "unbalanced" sample composition. For example, an SRS of $n = 200$ university students might contain substantially more women (e.g., 130) than men (e.g., 70), even if the population is evenly split by gender.
  - This is not a theoretical problem, but the results may appear more acceptable to a general audience if the subsamples are "balanced."

- In practice, stratified sampling requires that **stratum membership is known before sampling**.

  $\Rightarrow$ Stratification may therefore be theoretically desirable but infeasible if the information needed to assign units to strata is **not available on the sampling frame**.

- If **response rates** are expected to differ across strata, the initial sample sizes n_h may be adjusted accordingly. Strata with lower expected response rates may, for example, be sampled more heavily.

- If only the stratum population sizes N_h, $h = 1, \ldots, H$, are known, but unit-level stratum membership is unavailable before sampling, **post-stratification** may be used instead.

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